Sensor Calibration Checklist

Synthetic reference document - AI Factory Intelligence Command Center. For demonstration only; not a real operating procedure.

1.0 Scope

This checklist covers periodic calibration of the sensors that feed the predictive models: air and process temperature, rotational speed, torque, and the tool-wear timer. Perform at every preventive maintenance interval and after any sensor replacement.

2.0 Pre-Calibration Checks

2.1 Machine at ambient, powered on for at least 30 minutes for thermal stability.

2.2 Reference instruments in date: calibrated RTD, optical tachometer, reference torque cell.

2.3 Record ambient air temperature; it should read 295 - 305 K.

3.0 Temperature Sensor Calibration

3.1 Compare the air-temperature RTD against the reference at two points. Acceptance: within 0.5 K.

3.2 Compare process-temperature RTD similarly. Acceptance: within 0.5 K.

3.3 Verify the derived air-to-process difference reads at least 8.6 K under a light test load. A smaller reading with low rpm is the HDF warning band - investigate before returning to service.

4.0 Rotational Speed Calibration

4.1 Command 1500 rpm and measure with the optical tachometer. Acceptance: within 15 rpm.

4.2 Repeat at 2500 rpm. Acceptance: within 25 rpm.

4.3 Confirm the controller flags speeds below 1380 rpm as an HDF-relevant operating region.

5.0 Torque Sensor Calibration

5.1 Apply known loads with the reference torque cell at 10, 30 and 50 Nm. Acceptance: within 1 Nm or 2 percent, whichever is larger.

5.2 Check that the overstrain calculation (tool wear times torque) uses the corrected torque value.

5.3 Verify the Type-specific OSF thresholds are configured: L 11000, M 12000, H 13000 min*Nm.

6.0 Tool-Wear Timer Verification

6.1 Confirm the cumulative wear counter increments only during spindle-on cutting time.

6.2 Verify the 200-minute change reminder and the 240-minute hard stop are both active.

6.3 Reset the counter only when a tool change is logged with a tool ID.

7.0 Power Calculation Check

Confirm mechanical power is computed as torque times angular velocity and that the PWF band of 3.5 - 9 kW is configured. Simulate an out-of-band value and confirm the alert fires.

8.0 Sign-Off

Record every measured value, pass/fail, the technician, and the approving supervisor. Any FAIL requires a repeat calibration after correction. File the completed checklist with the machine maintenance record.